



Experiment-5

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Semester: 5th
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Date: 11/08/2026
Subject Code: 24CSH-337

1. Aim/Overview of the practical:

To configure and understand Dynamic Routing using the RIP (Routing Information Protocol) and to check end-to-end connectivity across multiple networks using Cisco Packet Tracer.

2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

3. Theory:

A) What do we mean by Routing?

Routing refers to the mechanism by which a router decides the most suitable path to move a data packet from its source network towards the destination network. Whenever the source and destination devices are placed on two different networks, the intermediate routers rely on their routing tables to figure out where the packet should be forwarded next.

B) What is meant by Dynamic Routing?

Dynamic Routing is an approach in which routers build and update their own routing tables on the fly by exchanging information with neighbouring routers, instead of the administrator having to type in every route by hand.

Benefits of Dynamic Routing

Benefit	Description
Self-learning	Routers discover reachable networks on their own.
Reduced admin effort	No need to type each route manually.
Resilience	An alternate path is picked automatically if a link fails.
Grows well	Works comfortably even as the network expands.

Commonly Used Dynamic Routing Protocols

Protocol	Stands For
RIP	Routing Information Protocol
OSPF	Open Shortest Path First
EIGRP	Enhanced Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

C) RIP (Routing Information Protocol)

RIP is a distance-vector style routing protocol that lets routers share reachability information with directly connected neighbours automatically. It is regarded as one of the earliest and most straightforward dynamic routing protocols in use.

Why was RIP introduced in the first place?

As networks grow, keeping every static route updated by hand becomes impractical. RIP was designed to solve exactly this problem by letting routers:

1. Discover neighbouring networks on their own.
2. Keep their routing tables current without manual edits.
3. Pick the best available route automatically.

How RIP actually works

RIP relies on Hop Count as its metric for choosing the best route — a hop being simply one router that a packet has to pass through on the way to its destination.

Example: R1 → R2 → R3 → Network X. Counted from R1, this path has a hop count of 2, since the packet passes through R2 and R3 before arriving.

RIP Hop Count Reference

Hop Count	Interpretation
1	Network reachable via one router
2	Network reachable via two routers
5	Network reachable via five routers
15	Farthest network RIP can still reach
16	Treated as unreachable (infinity metric)

Sample RIP Routing Table

Destination Network	Next Hop	Hop Count
172.16.1.0	Directly Connected	0
172.16.2.0	10.0.12.2	1
172.16.3.0	10.0.12.2	2

Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Setup	Entered manually by the network admin.	Learned automatically via a routing protocol.
Updates on change	Admin must edit routes by hand.	Table refreshes itself when topology changes.
Best fit for	Small, rarely-changing networks.	Medium/large, frequently changing networks.
Router CPU load	Minimal – no path computation.	Higher – continuous route computation.
Router memory use	Minimal.	Higher, to hold learned routes.
Link bandwidth use	None used for updates.	Some, due to periodic advertisements.
Security	Slightly safer – nothing advertised.	Slightly riskier – routes are advertised.

Feature	Static Routing	Dynamic Routing
Upkeep effort	Grows harder as network grows.	Stays manageable even as network grows.

4. Steps for Experiment/Practical:

A) Dynamic Routing with 6 PCs, 3 Routers and 3 Switches

1. Launch Cisco Packet Tracer and place 3 Routers, 3 Switches, and 6 PCs on the workspace.
2. Cable the devices together to match the intended topology.
3. Assign an IP address to every PC and to each router interface.
4. Set the correct Default Gateway on every PC.
5. Enable RIP on all three routers and advertise the directly connected networks.
6. Check the routing table on each router with: show ip route
7. Use the ping command to confirm connectivity across the different LANs.

B) Dynamic Routing with 8 PCs, 4 Routers and 4 Switches

1. Launch Cisco Packet Tracer and place 4 Routers, 4 Switches, and 8 PCs on the workspace.
2. Cable the devices together to match the intended topology.
3. Assign an IP address to every PC and to each router interface.
4. Set the correct Default Gateway on every PC.
5. Enable RIP on all four routers and advertise the directly connected networks.
6. Check the routing table on each router with: show ip route
7. Use the ping command to confirm connectivity across the different LANs.

5. Output:

Dynamic Routing using 6 PCs, 3 Routers and 3 Switches

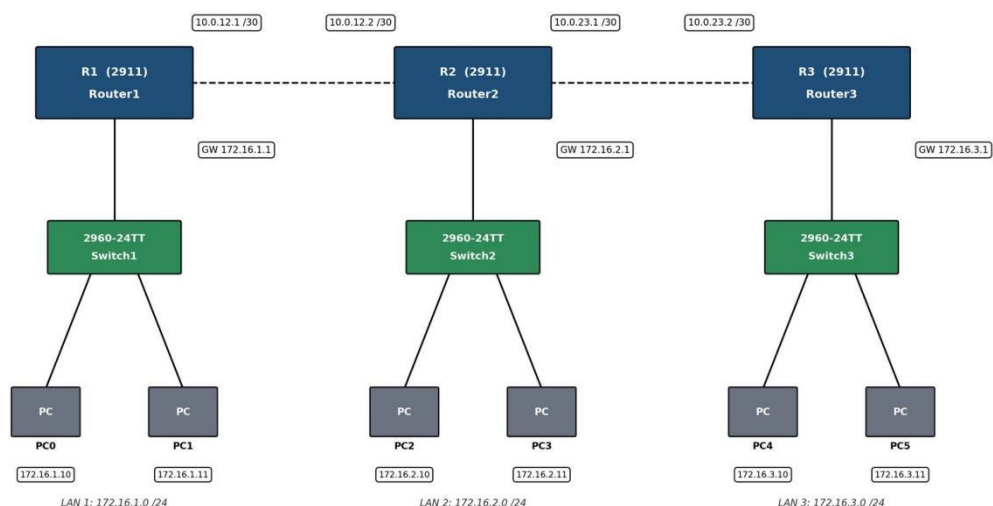


Fig 5.1 – Topology for Part A: 3 Routers, 3 Switches, 6 PCs (RIP-enabled)

Dynamic Routing using 8 PCs, 4 Routers and 4 Switches

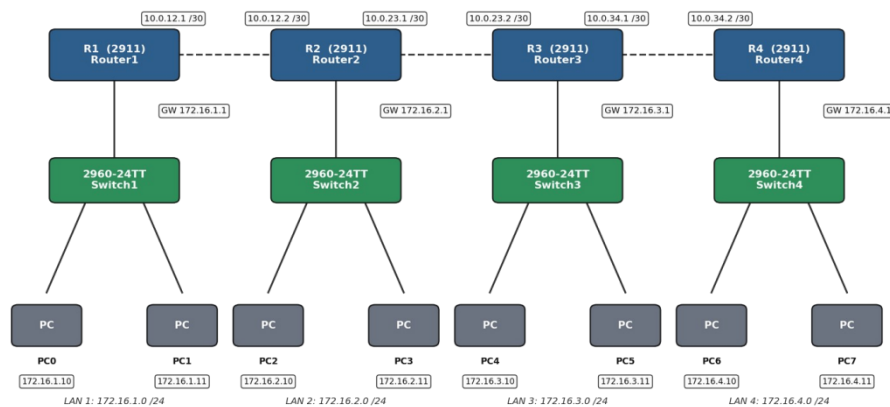


Fig 5.2 – Topology for Part B: 4 Routers, 4 Switches, 8 PCs (RIP-enabled)

Sample RIP configuration (Router1, Part A)

```
Router1 (config) # router rip
Router1 (config-router) # version 2
Router1 (config-router) # network 172.16.1.0
Router1 (config-router) # network 10.0.12.0
Router1 (config-router) # no auto-summary
```

6. Result

Dynamic Routing using RIP was configured and tested successfully in Cisco Packet Tracer for both the 3-router and 4-router topologies. Each router automatically learned the remaining networks through RIP, the routing tables populated correctly, and ping tests confirmed full connectivity between PCs placed on different LANs.

7. Learning Outcomes

- Understood what routing is and why it matters in a multi-network setup.
- Compared static routing against dynamic routing and identified when each is appropriate.
- Learned how RIP discovers and advertises routes between neighbouring routers.
- Understood how RIP relies on hop count to pick the best available path.
- Configured RIP on multiple routers and observed automatic route learning in action.
- Practiced verifying routing tables using the show ip route command.
- Verified inter-network connectivity between PCs using the ping command.